Plato/Lindberg

Lecturer:

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HSS, CUHK(SZ)

About the Course

Human Exploration of the Physical Universe

- Plato, Lindberg
- Aristotle, Lindberg
- Cohen, Newton

Human Exploration of the World of Life

- Darwin
- Watson
- Carson

Our Understanding of Human Understanding

- Poincare
- Kandel
- Needham
- Sivin, Shen Kuo
- Dunhan, Euclid

Why starting with Plato?

- Ontology (本体论)

  what's the nature of being, becoming, existence or reality?

了解科学知识

- Epistemology (认识论)

  what's the nature of knowledge, justification and the reality of belief?

了解科学的研究过程和方法

- Ethics (伦理学)

  what should we do, usually in terms of rights, obligations, benefits to society, fairness, or specific virtues?

了解科学技术对社会和个人所产生的影响

Lecture 1

Plato/Lindberg

Text 1a: Plato, *Republic*
Text 1b: David Lindberg, *The Beginnings of Western Science*

Core Questions:

- What is true knowledge?
- What is the purpose of education?

Aims:

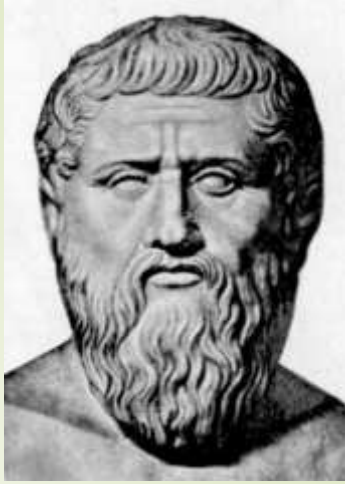
- How are two core questions related?
- How is Plato related to Western Science?

Outline

- Plato's view on reality
  - Allegory of the cave
  - World of forms
- Background of Plato's view
  - Influence from the outside
- The influence of Plato on modern science
  - Reason & sensible experience
  - Idealization

Profile of Plato

- Classical Greek **philosopher** and **mathematician.**
- Socrates + Plato + Aristotle: laid the foundations of Western philosophy.
- Great influence on **ethics** and **science.**



Plato: broadness

c.428 BC – *c*. 348 BC

Plato made foundational contributions in most of the areas of philosophy.

Plato defined many contemporary issues in philosophy.

Metaphysics: What is real?

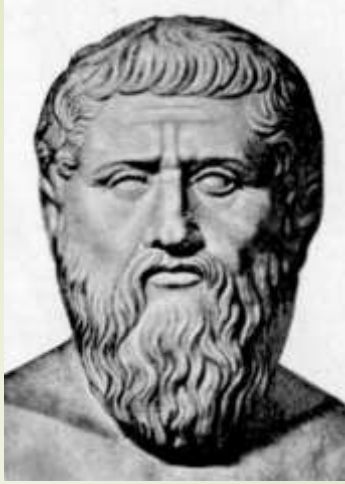
Epistemology: What is knowing?

Ethics: What is the good life?

Political philosophy: What is the best regime?

Aesthetics: What is beauty?

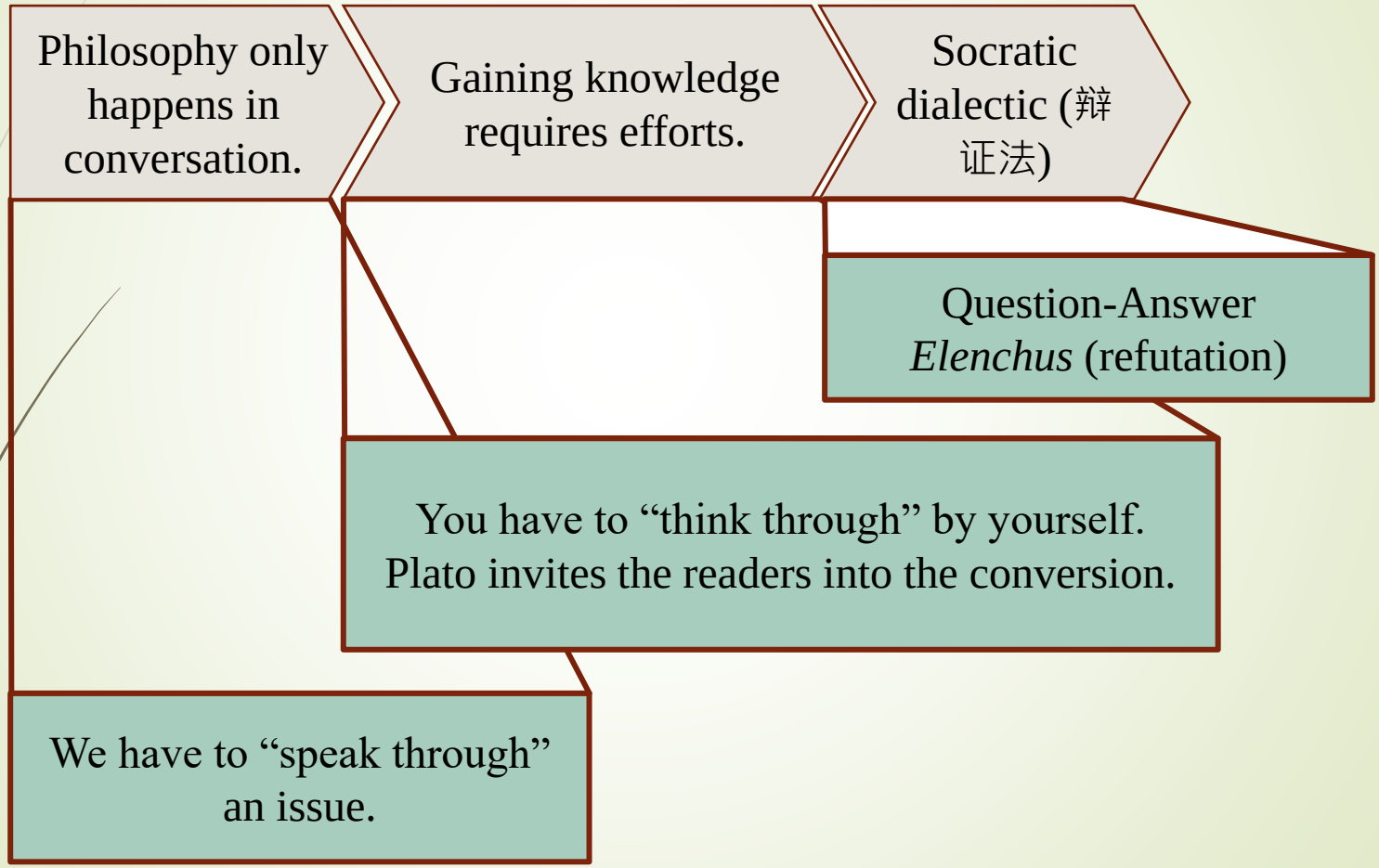
Profile of Plato



Plato: broadness
c.428 BC – *c*. 348 BC

- Classical Greek **philosopher** and **mathematician.**
- Socrates + Plato + Aristotle: laid the foundations of Western philosophy.
- Great influence on **ethics** and **science.**
- Founded Academy (387 BC), including subjects of philosophy, math and gymnastics.
- Notable works (In dialogues)
  - Republic (理想国)
  - Phaedo (斐多篇)
  - Symposium (会饮篇)
  - Meno (美诺篇)
  - …

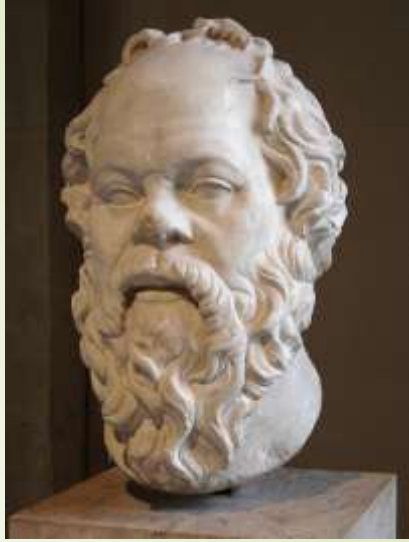
Teaching in dialogues



Example of Socrates Dialectic

- S(Socrates): What are fish?
- P(Pupil): Fish are animals which live in the sea.
- S: Do crocodiles (鳄鱼) live in the sea? Are crocodiles fish?
- P: No, crocodiles are not fish.
- S: So can you tell me what fish are?
- P: Fish are the animals which live in the sea and have no limbs (四肢).
- S: Do dolphins live in the sea and have no limbs? Are they fish?
- P: No, dolphins are not fish.
- S: So can you tell me what fish are?
- P: Fish are…

Socrates



Socrates
(c. 470 BC-399 BC)

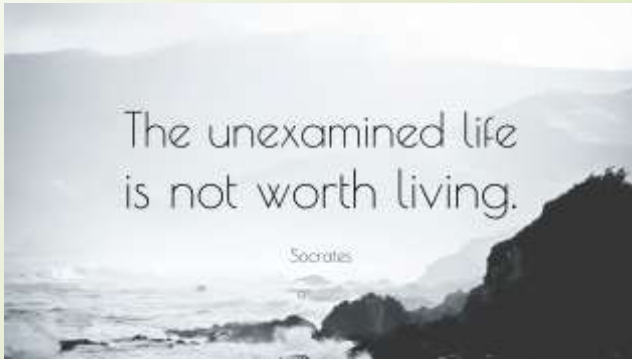
- Recognized as the first Greek philosopher to put an emphasis on moral problems. Connection with values in science: *e.g.* honesty, humility, *etc*...

- Dialogical method

  (Socratic *elenchus* (refutation))

  - Galileo's "*Dialogue on two chief systems of the world* " seems inspired to Socrates' argumentation style.

- Died of critic of democracy (民主批判)



Death of Socrates

Reason: “impiety (不虔诚)” & “corrupting the young”



Socrates “raised the cup to his lips and very cheerfully and quietly drained it.”

Republic (Totally 10 books)



Title page of the oldest complete manuscript (late 9th century)

- Best-known work, one of the world's most influential works of philosophy and political theory, both intellectually and historically.
- "What is justice?"
- Personal justice ⇔ justice in a city
- The ruler should be a philosopher.
  - He should know the Form of the Good.

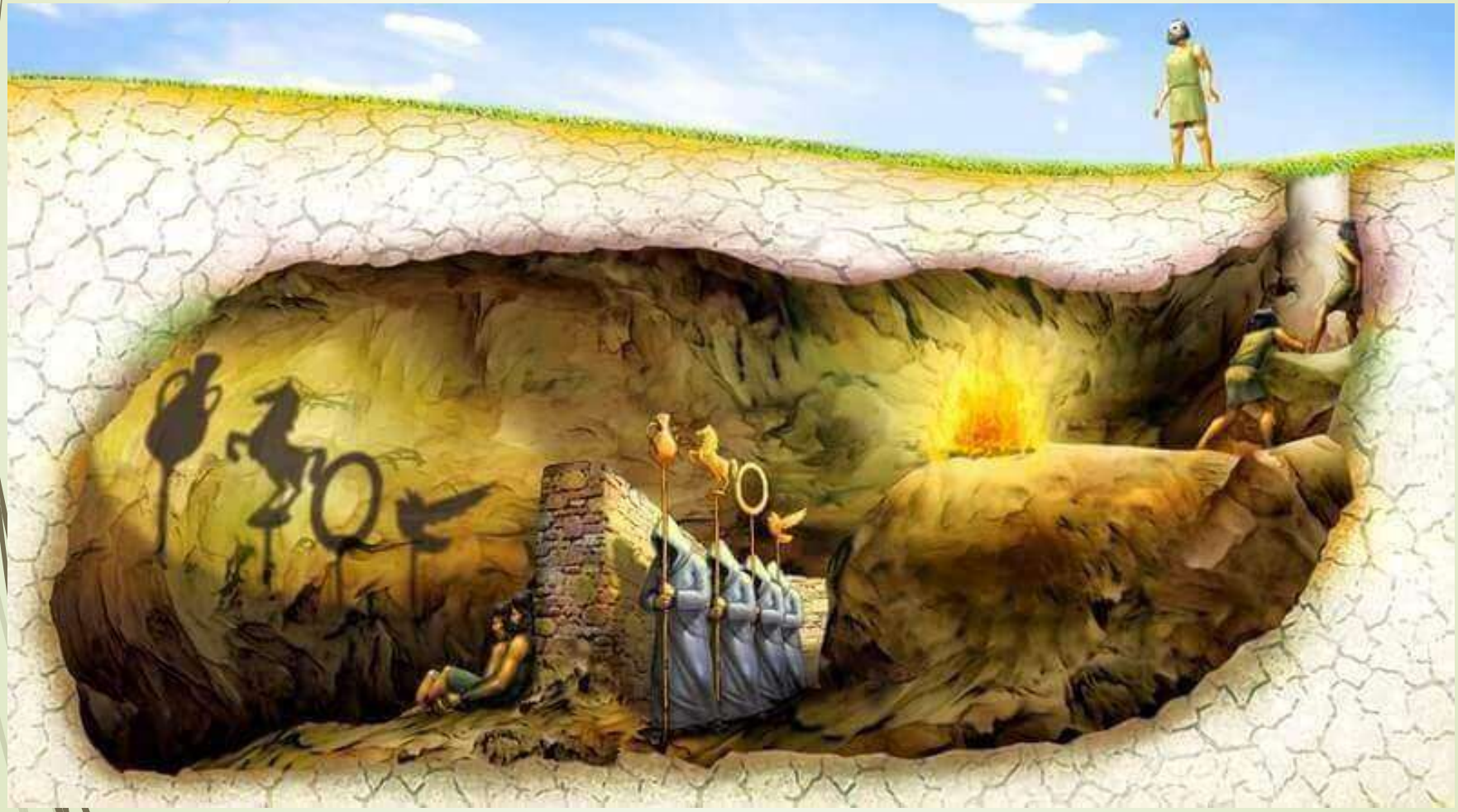
Book I-V	the Utopia portion, portraying the ideal community, starting from an attempt to define justice
Book VI-VII	since philosophers are seen as the ideal rulers of such community, this part of the text concentrates on defining precisely what a philosopher is
Book VIII-X	discusses several practical forms of government, their pros and cons.

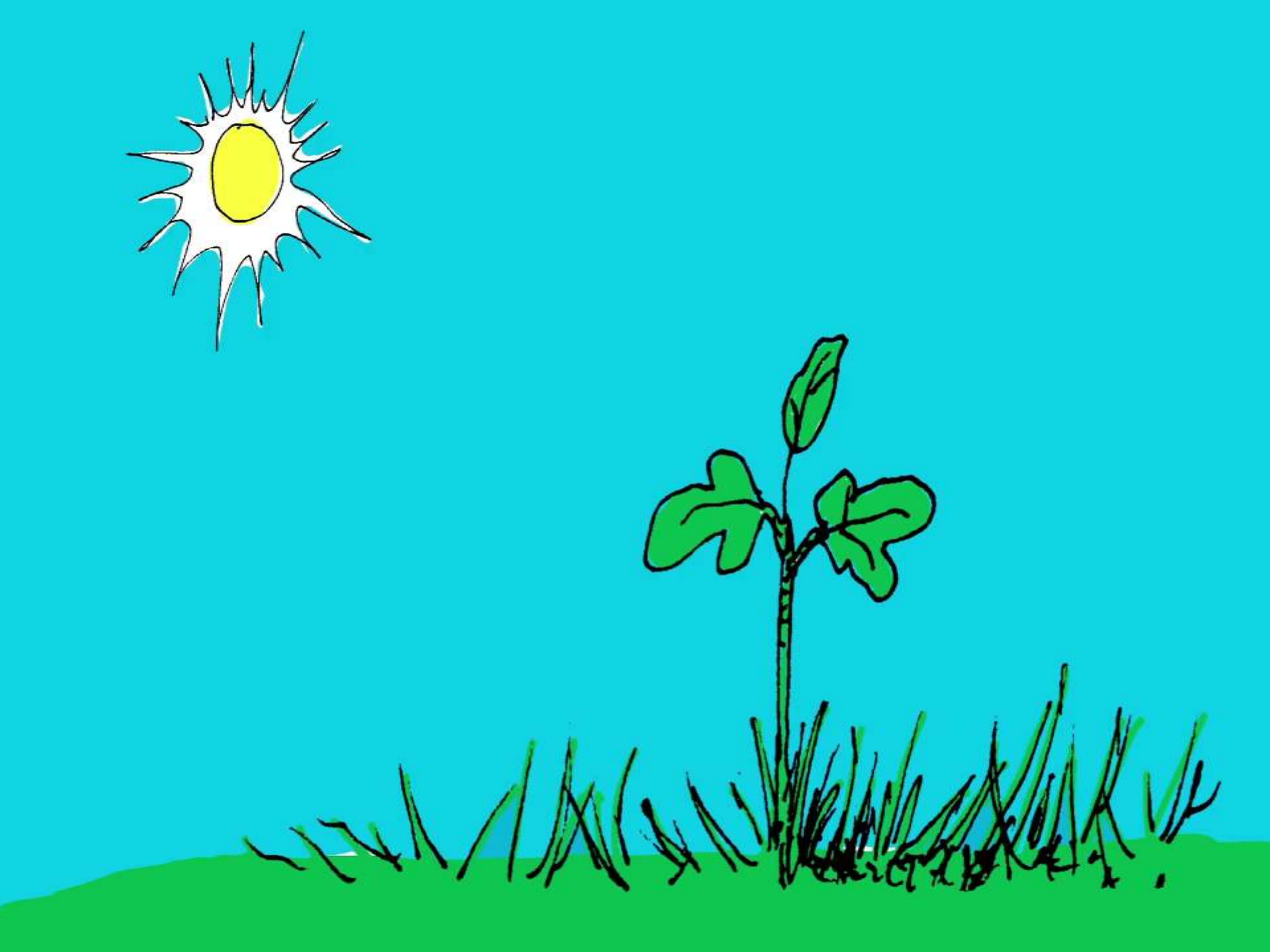
http://en.wikipedia.org/wiki/The_Republic_(Plato)

Book VII

- What is a philosopher?
- What is reality?
- How to learn?

Allegory of the Cave

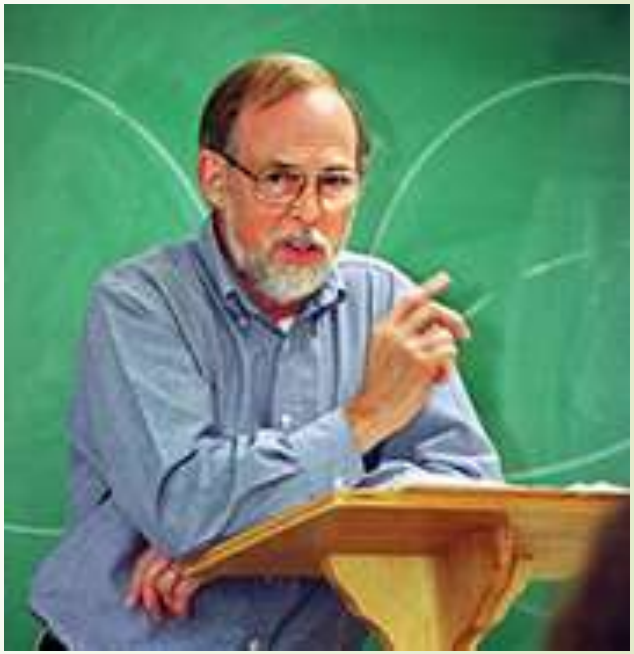




David C. Lindberg

- American historian of science
- Main focus:

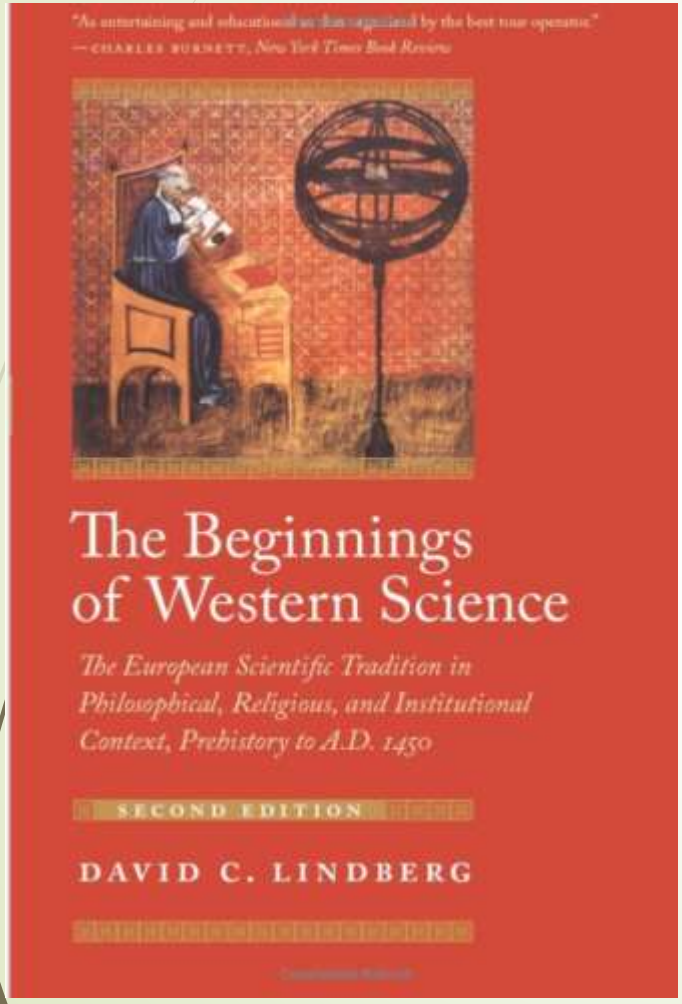
  History of medieval and early modern science, especially physical science and the relationship between religion and science.



David C. Lindberg (1935-2015)

http://histsci.wisc.edu/people/faculty/lindberg/lindberg.shtml

Text 1b: The Beginnings of Western Science



- A textbook for the history of science
- A guide for how Plato's view shaped the way of thinking of natural philosophers
- Chapter 2: Introducing Plato's World of Forms

Plato's two realms

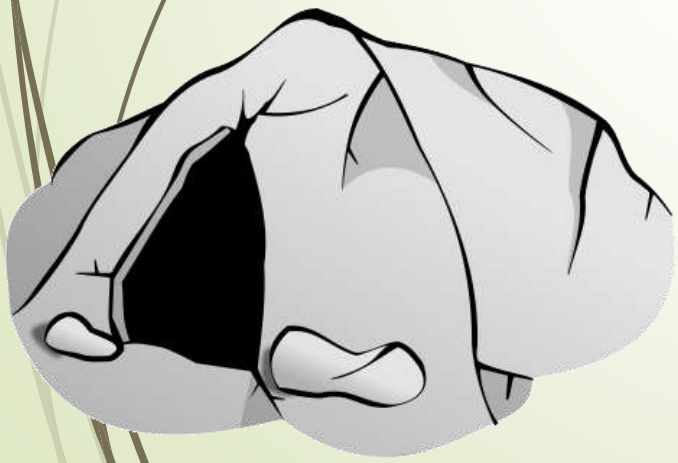
Appearance ≠ True reality



True reality

World of Forms/Intelligible world (理型世界）
Abstract, changeless, absolute.

- Outside the cave



Appearance

Sensible/material world (物质世界）

- Inside the cave

Text 1b: Lindberg

Plato's View on Reality

- What is true knowledge?

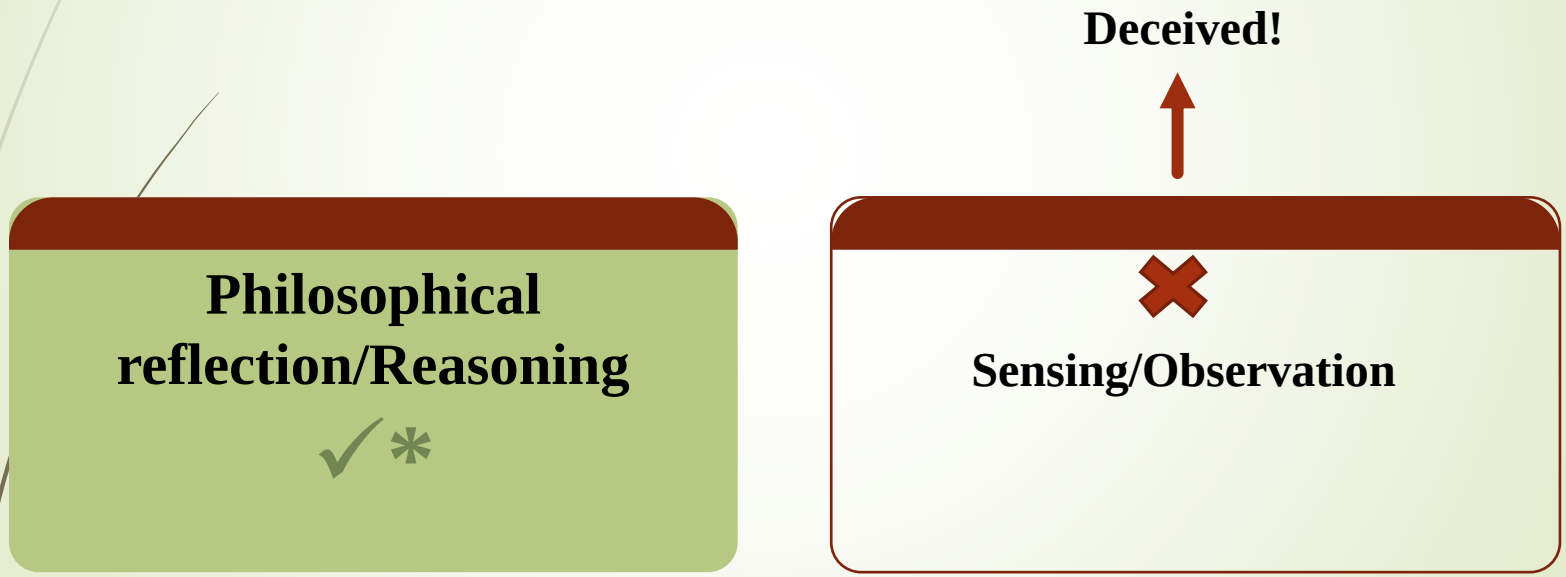
World of Forms /Intelligible world

- Perfect
- Incorporeal, intangible, insensible
- Changeless

Sensible/material world

- Imperfect
- Corporeal
- Changeable

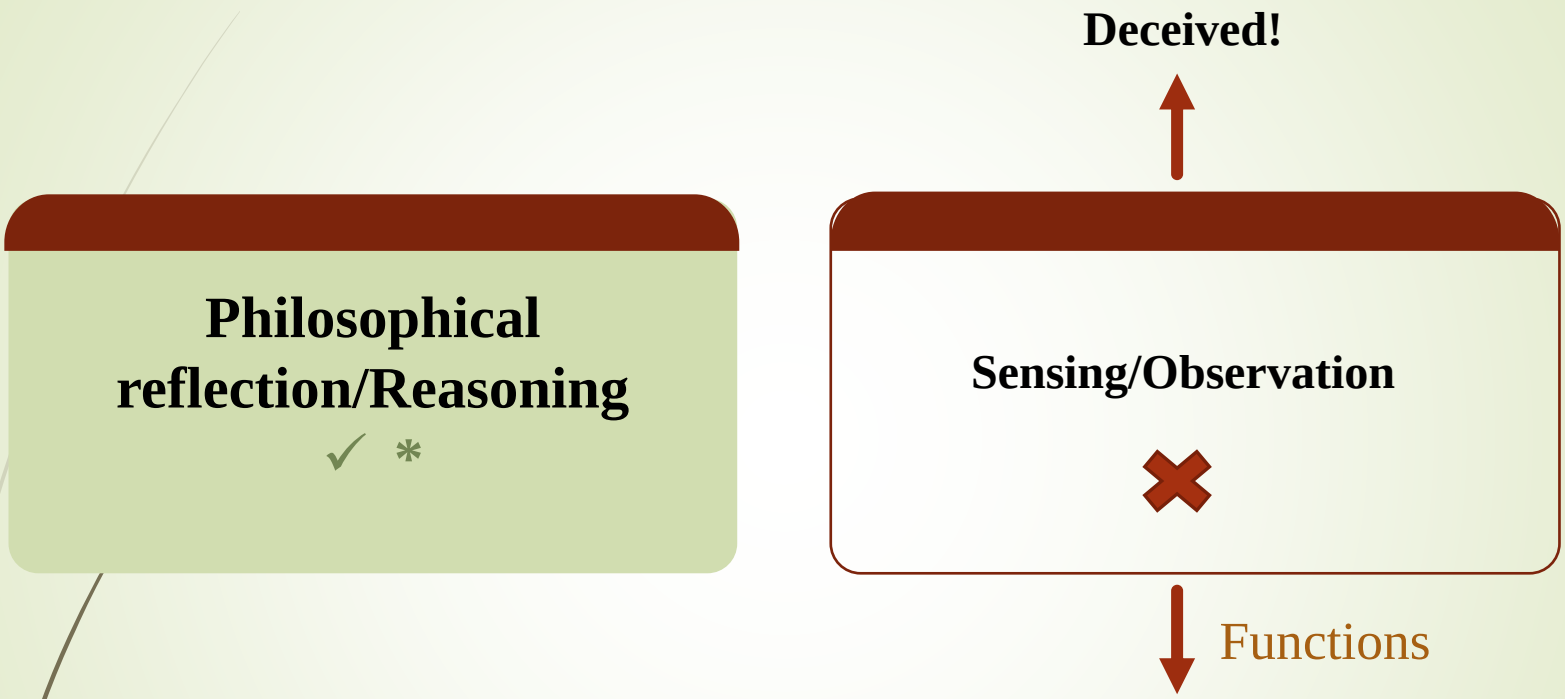
How to get true knowledge?



An example of the optical illusions



How to get true knowledge?



1. Sense experience may provide wholesome recreation.
2. Observation of certain sensible objects may serve to direct the soul toward nobler objects in the realm of forms.
3. Plato argued (in his theory of reminiscence) that sense experience may actually stir the memory and remind the soul of forms that it knew in a prior existence.

Theory of reminiscence (回忆论)

- This theory is based on the assumption of the **immortality** of the soul. So it holds all knowledge. Human soul exists beyond the physical world as an undying entity which both precedes and outlives the human body.
- In essence, time and learning are illusions because all knowledge and all events are already contained in memory. Reality is merely the unfolding process of remembering.

Theory of Reminiscence (回忆论)

- Theory of Reminiscence

  We have the knowledge

  Our senses and experience just stir our memory

  Our souls are reminded of the form and “remember” the knowledge

- We lived in the world of form. There was the perfect desk.
- We now live in the material world. There are material desks. A material desk can remind us of the perfect desk.
- We therefore know it is a desk.

Déjà vu?

This is a French phrase translating literally to "already seen". The feeling that one has lived through the present situation before.

Text 1b: Lindberg

Plato's View on Reality

- What is real/true knowledge?

World of Forms/ Intelligible world (理型世界)	Sensible/ material world (物质世界)

- How to get true knowledge? Why?

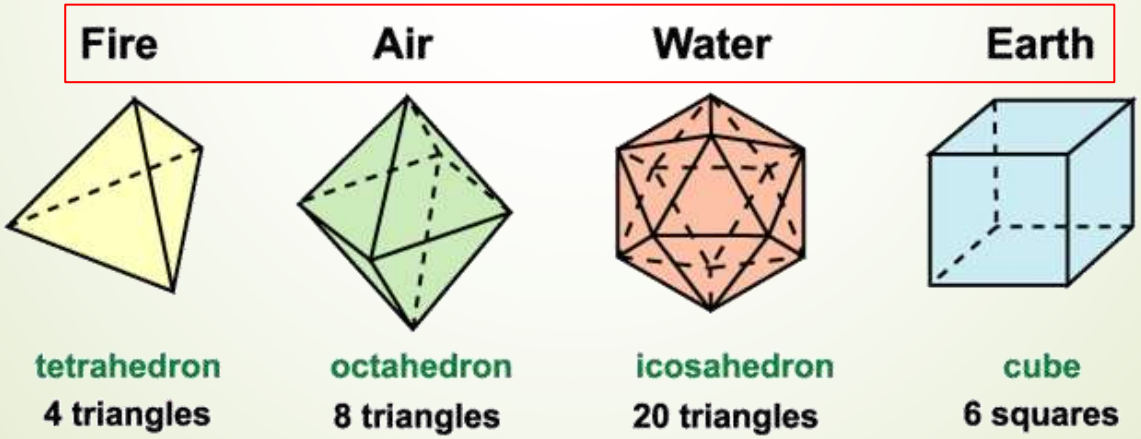
Philosophical reflection/Reasoning	Sensing/ Observation

- How to cultivate philosophical reflection/reasoning?

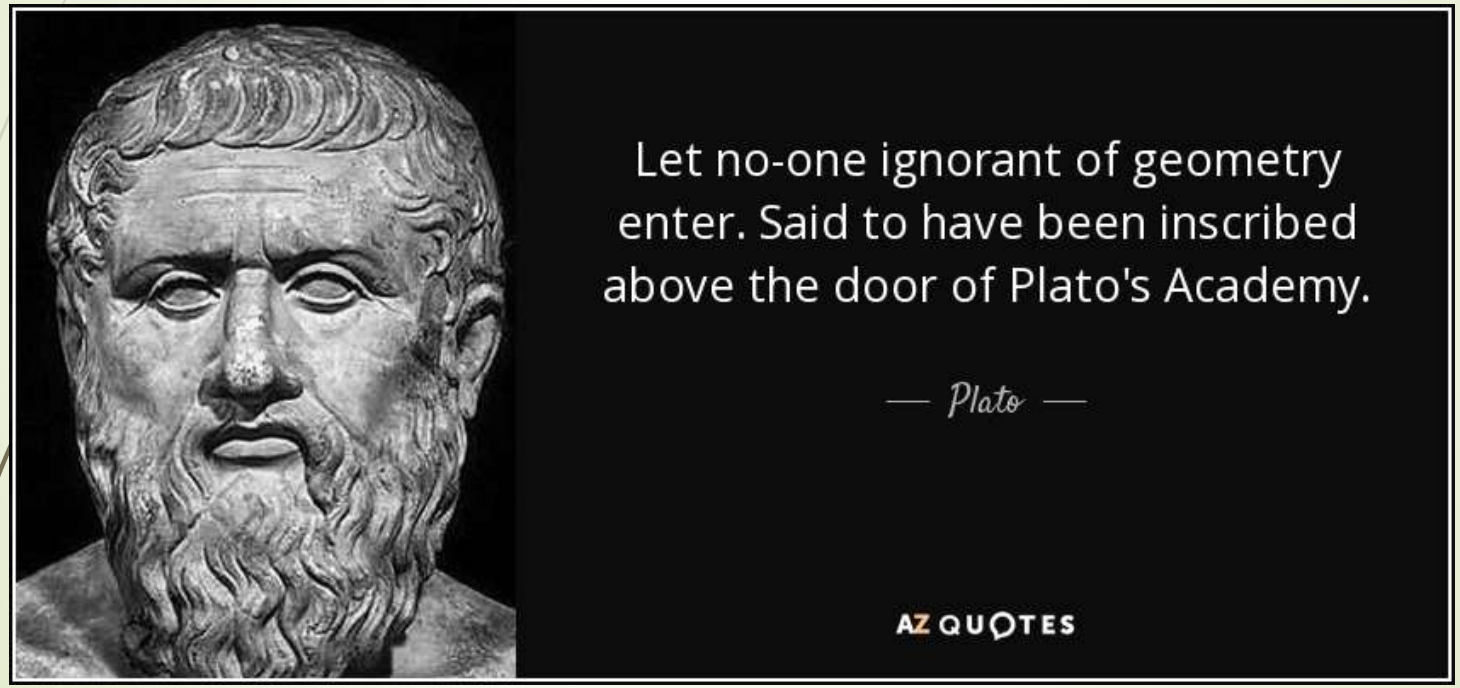
Geometry

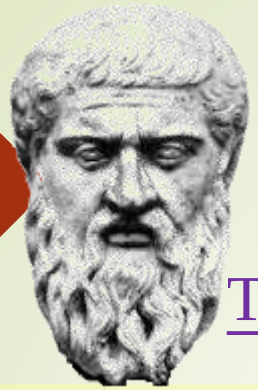
Platonic solids (柏拉图立体)

- Four elements: Fire, Air, Water, Earth
  -proposed by Empedocles (pre-Socratic philosopher)
- Four elements can be reduced to something more fundamental.
  -proposed by Plato



- At the entrance of Academy was the inscription (题词):

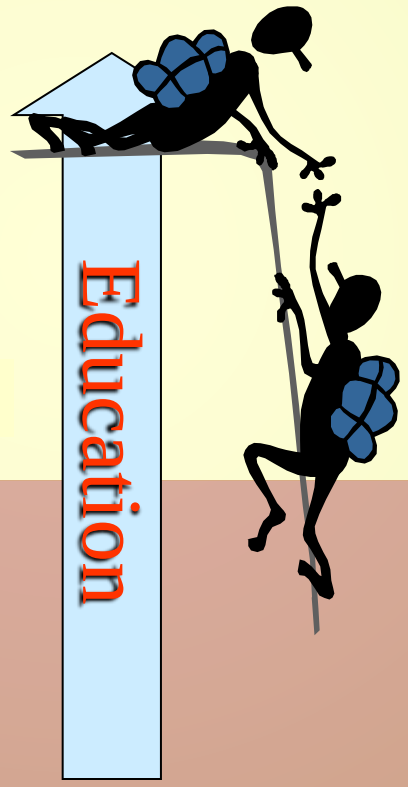




Education is…

Epistemological
Text 1b para 34

Type of object		Type of Knowledge
Sun	**Intelligible realm**	Philosophical reflection
Things in the sky, the light of the stars and the moon.	$a^2 + b^2 = c^2$	Mathematical reasoning
Physical objects: artifacts, statues of people and other animals.	**Sensible realm**	Beliefs about physical things
Shadows of the artifacts and statues, *etc*.		Sensory experience, illusions



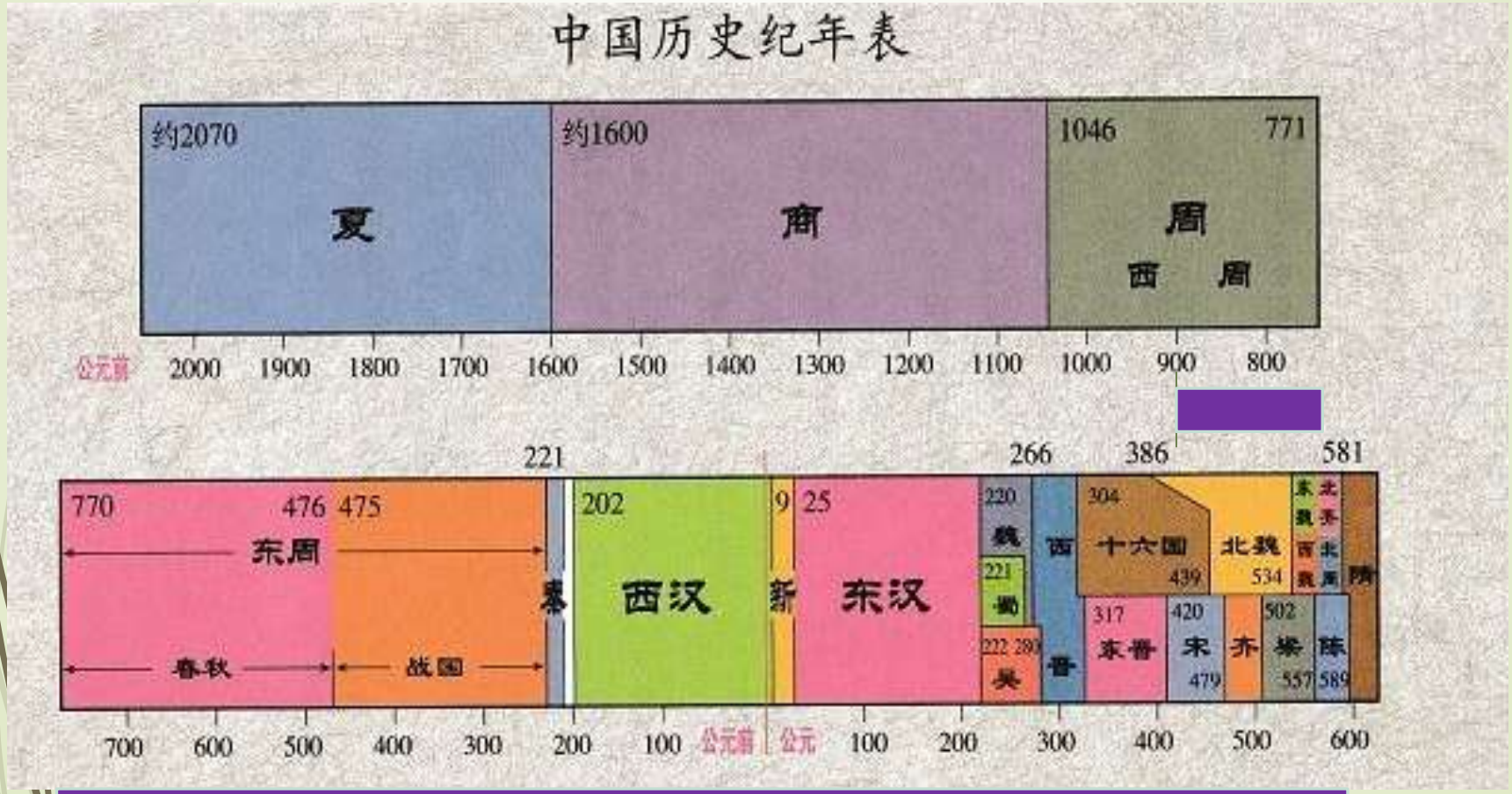
http://en.wikipedia.org/wiki/Analogy_of_the_divided_line

Outline

- Plato's view on reality
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Ancient Greece

-9th century BC to 6th century



Greek Mythology (神话)

- Sun and Moon as deities (神明)
- Natural phenomena as mighty feats willed by god
- **Capricious (变幻莫测的), non-predictable world**



Ancient Greek Gods and Goddesses

Ancient China Mythology







The beginning of ancient Greek philosophy

Greek Mythology:

- Sun and Moon as deities
- Natural phenomena as mighty feats willed by god.
- **Capricious, non-predictable world**

Ancient Greek Gods and Goddesses
Homer: Iliad, Odyssey

Early in the 6th century BC:

- Not personify nature (exclude God)
- Things behave according to their natures
- **Search for orderly, predictable world**

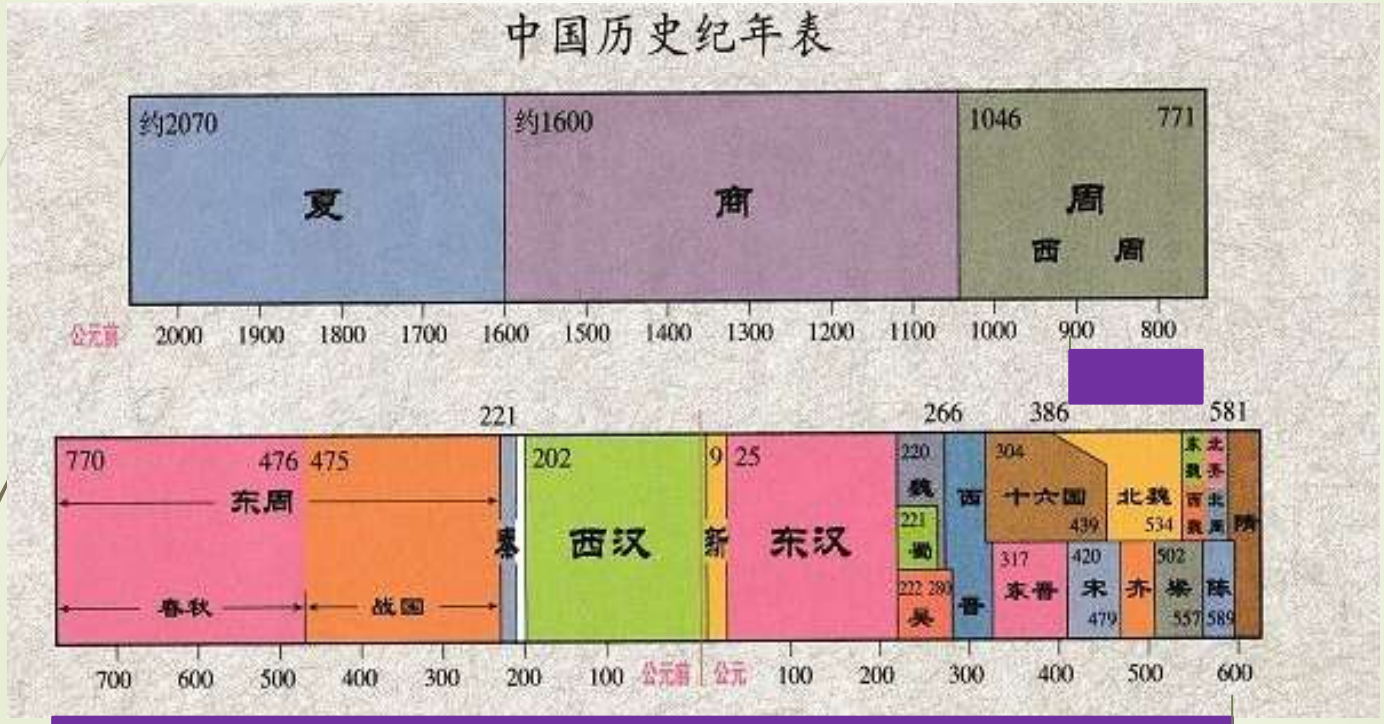
Philosophy and philosophers:

- Philosopher: a person who seeks wisdom or enlightenment.
- Thales, Parmenides, Heraclitus, Pythagoras, Socrates, Plato, Aristotle…

Reality & Change

Classical Greece

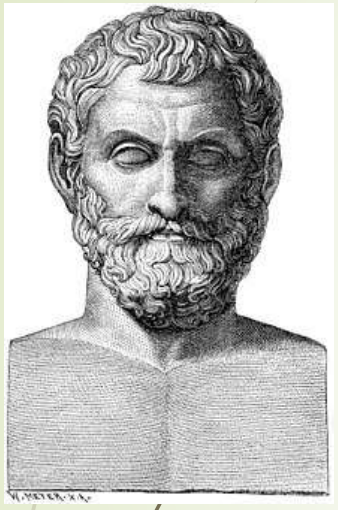
-6th century BC to 4th century BC



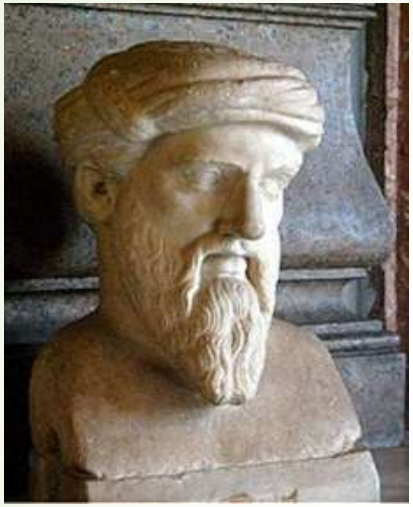
- ✓ Foundation of modern western culture
- ✓ The cradle of western civilization

Philosophers in Classical Greece

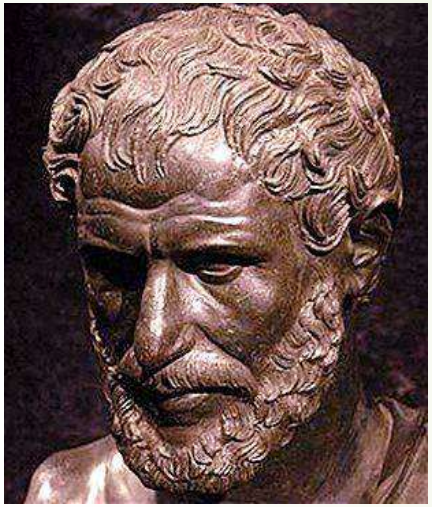




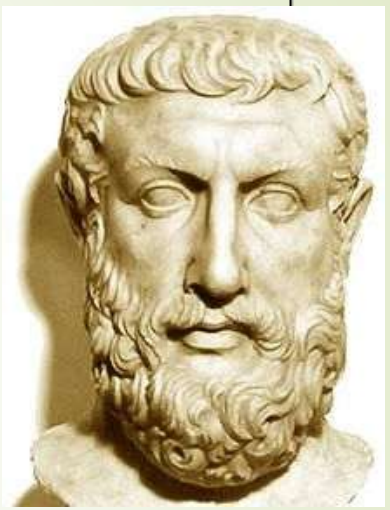
Thales
（泰利斯）
c. 624 - 546 BC



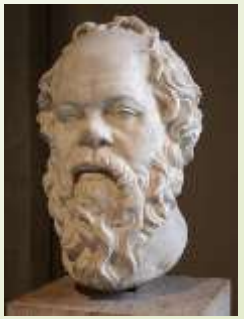
Pythagoras
（毕达哥拉斯）
570 - 495 BC



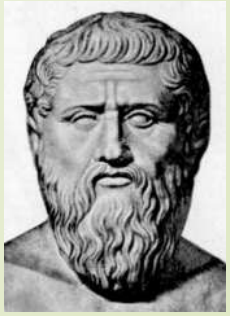
Heraclitus
（赫拉克利特）
535 - 475 BC



Parmenides
（巴门尼德）
515 - c.450 BC

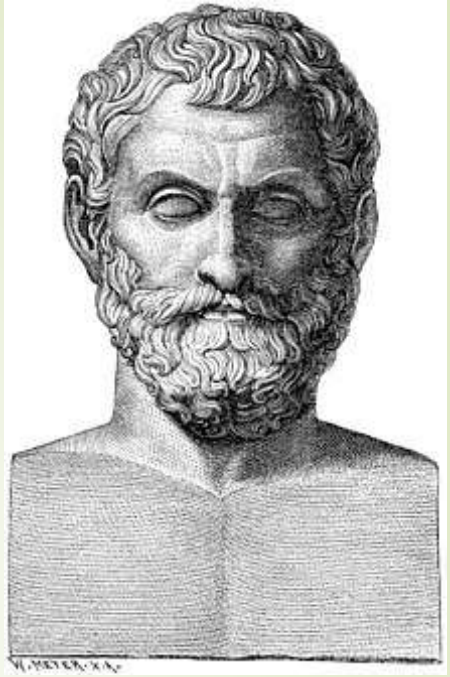


Socrates
470 - 399 BC



Plato
428 - 348 BC

Thales (泰勒斯)



Thales
(c. 624 – 546 BC)

First Philosopher in Greek tradition

- Early Spring and Autumn Era (春秋初期) in ancient China
- Nature is completely material, independent of supernatural beings

What is **matter** made of ?

Everything is originally made of **water.**

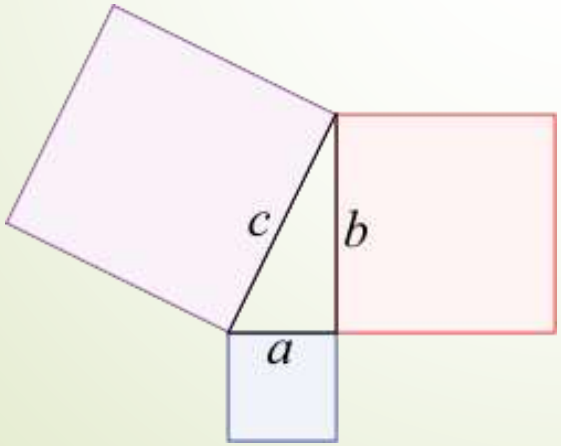
- The answer was nonsense
- But the question made good sense
- **Nature could be explained** without talking about gods

38 Pythagoras (毕达哥拉斯)



Pythagoras
(c. 570 – c. 495 BC)

- Greek philosopher, mathematician, founder of the religious movement called Pythagoreanism.
- Pythagoras' theorem (勾股定理).



$$a^2 + b^2 = c^2$$

39

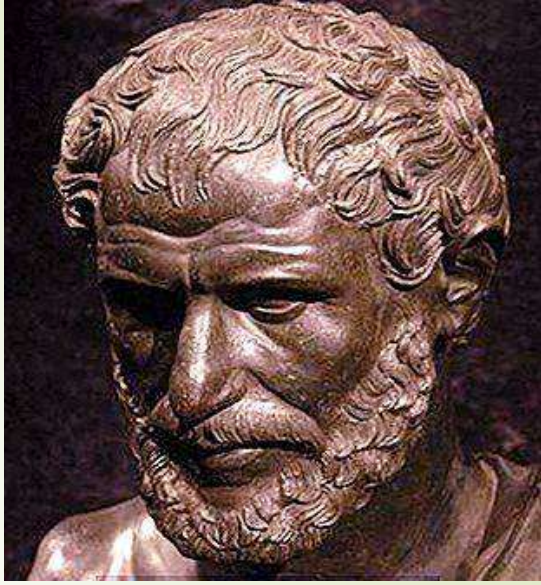
Pythagoras

- “Everything is number.” (positive integers)
- “Numbers are the only real thing.”
- “The elements of numbers to be the elements of all things.”
- “Number as a fundamental aspect of reality.”
- “The soul was immortal.”

40

Heraclitus (赫拉克利特)

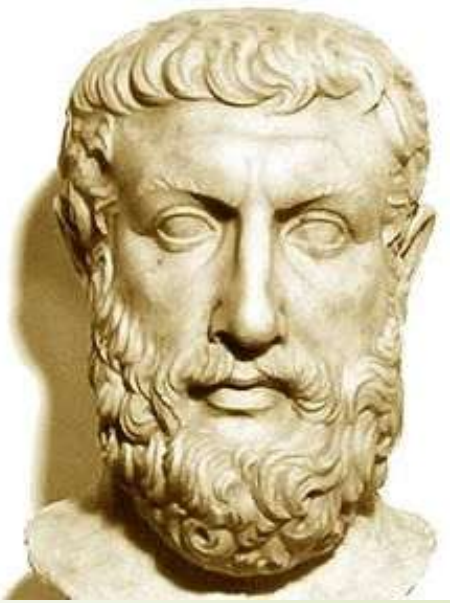
- There is only change.
- Everything is in flux.
- Our sensory faculty is reliable.



Heraclitus
(535 - 475 BC)



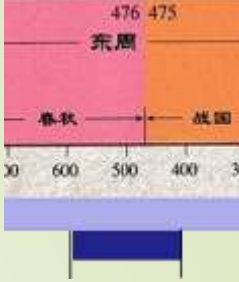
Parmenides (巴门尼德)

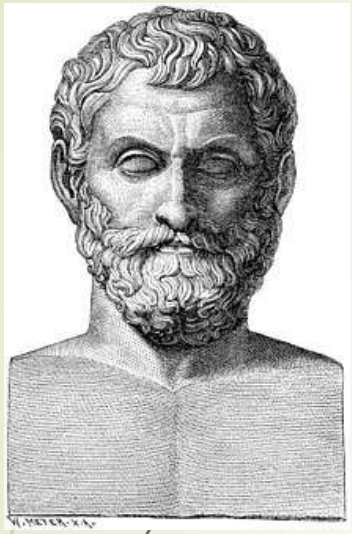


Parmenides
(515 - c.450 BC)

- Greek philosopher
- About reality & change
  - reality is one, change is impossible, existence is timeless, uniform, necessary, and unchanging
  - sensory faculties lead to conceptions which are false and deceitful
- Denying the possible existence of a void (empty space).

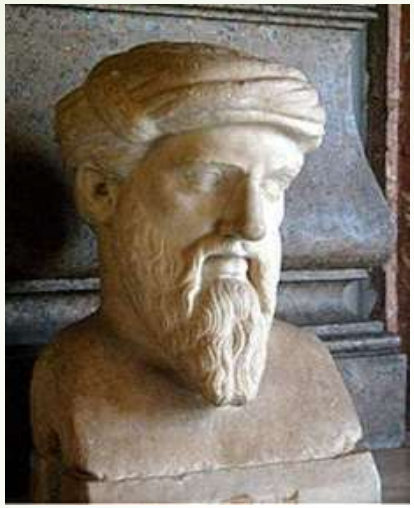
Philosophers in Classical Greece





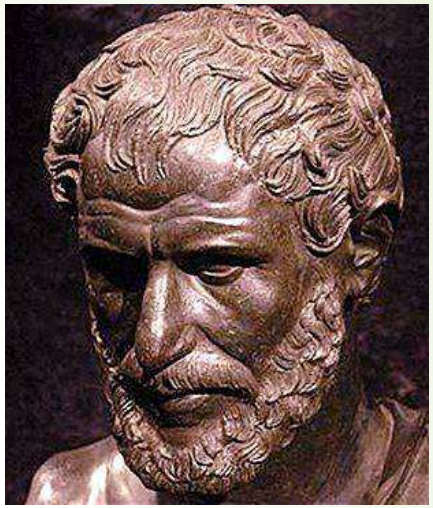
Thales

Everything is originally made of water.



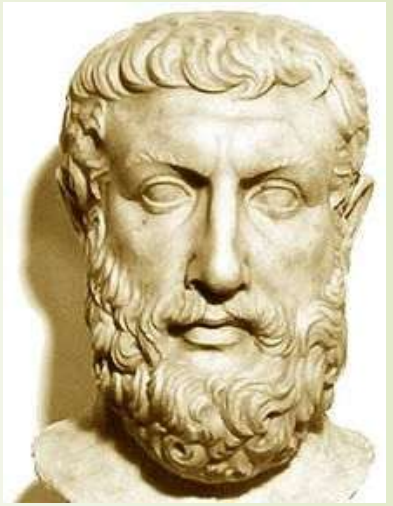
Pythagoras

Mathematics as a basic tool for investigating the reality.



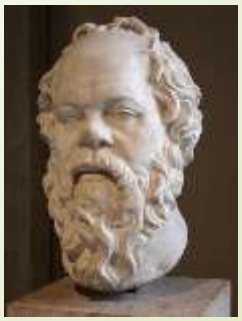
Heraclitus

There is only change.

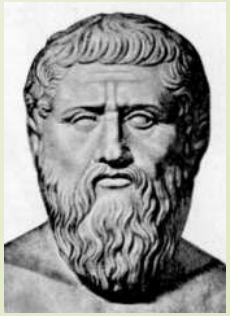


Parmenides

Change is impossible.



Socrates
470-399 BC

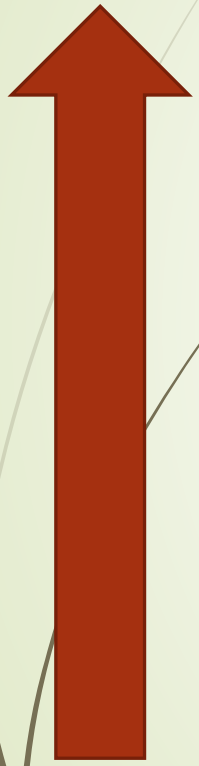


Plato
428-348 BC

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  - Influence from the outside
- The influence of Plato on modern science
  - Reason & sensible experience
  - Idealization (*e.g.* Newton's first law)

Plato's view



The world of forms → Perfect
Common properties

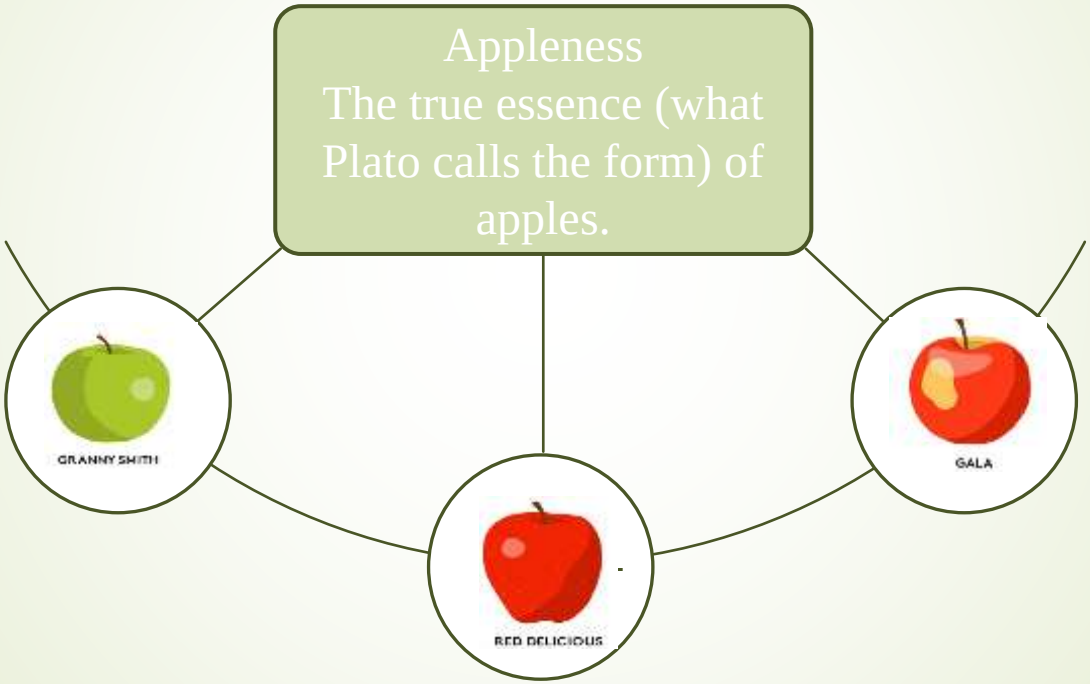
- Intelligence
- Reason
- Mind/soul

Visible world → Imperfect
Individual characteristics

- Sense
- Experience
- Body

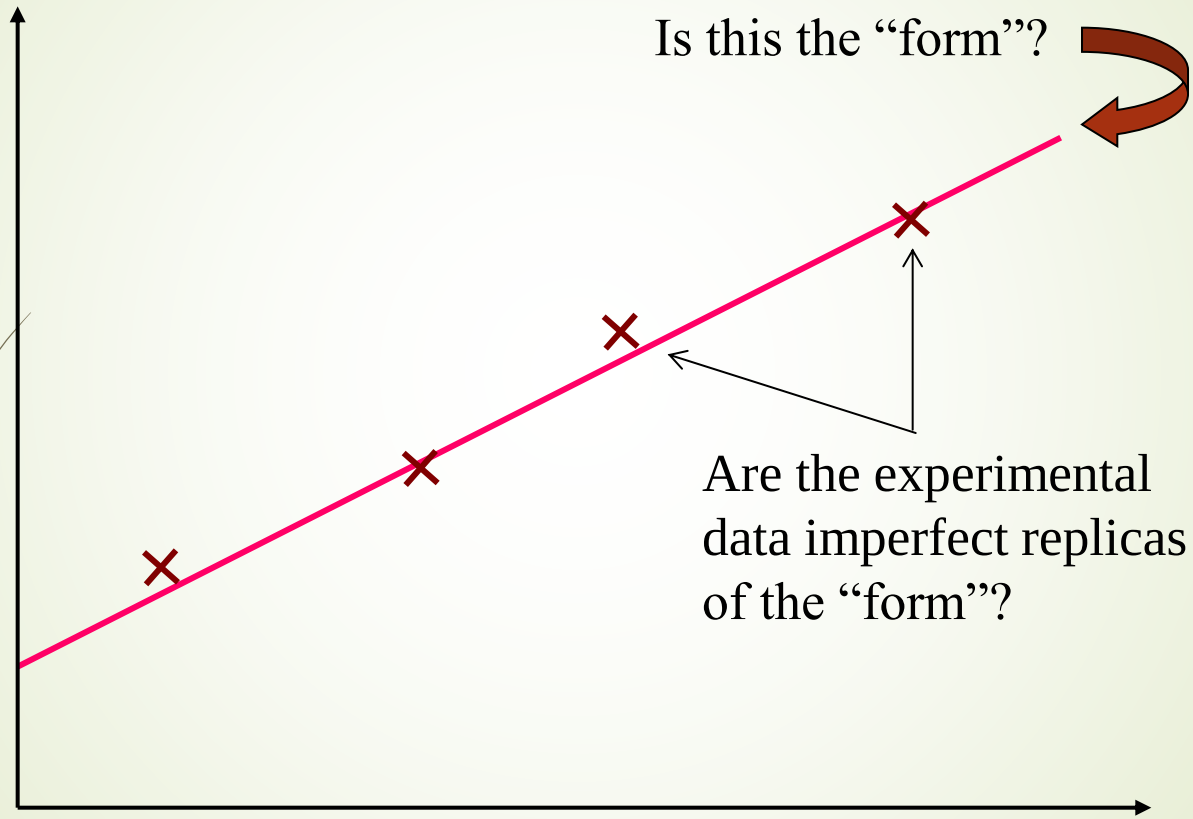
The World of Forms

"the cause of all that is correct and beautiful in anything"



- To gain true knowledge, we must set aside all characteristics peculiar to things as individuals and seek the shared characteristics that define them into classes.

Idealization: a prominent feature of modern science



- We develop models or laws that overlook the incidental in favor of the essential.

Plato's View & Modern Science

- Not denying the role of experiences.
- The form or idea is the common property.
  - Not characteristics as individuals but shared characteristics.
- This common property (the idea or form) has objective, independent, and indeed prior existence.

Outline

- Plato's view on reality
  - Allegory of the cave
  - World of forms
- Background of Plato's view
  - Influence from the outside
- The influence of Plato on modern science
  - Reason & sensible experience
  - Idealization

Reading assignment

-In Dialogue with Nature

- Text 1a: Plato's allegory of the cave
- Text 1b: David Lindberg's “ *Plato's World of Forms* ”